8. Minimax Coin Game

Python practical code

```
coins = [3, 9, 1, 2]

def minimax(c, turn):
    if not c:
        return 0

    if turn:
        return max(
            c[0] + minimax(c[1:], False),
            c[-1] + minimax(c[:-1], False)
        )
    else:
        return min(
            -c[0] + minimax(c[1:], True),
            -c[-1] + minimax(c[:-1], True)
        )

while coins:
    left = coins[0] + minimax(coins[1:], False)
    right = coins[-1] + minimax(coins[:-1], False)

    if left >= right:
        ai = coins.pop(0)
    else:
        ai = coins.pop()

    print("AI picked:", ai)

    if coins:
        print("Remaining:", coins)
        choice = input("Pick left or right: ")

        if choice == "left":
            human = coins.pop(0)
        else:
            human = coins.pop()

        print("Human picked:", human)
```